A-A-51141
31 October 1984

COMMERCIAL ITEM DESCRIPTION

PLATE, LAPPING

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-P-446.

This description covers one type of lapping plate. The lapping plates are intended to be used in lapping operations in ordinary machine shop practices to assure a perfect fit of matching parts.

Salient characteristics.

Classification. Lapping plates covered by this description shall be of the sizes listed in table I and shall conform to the dimensions and weights listed in table I for the size required. The size required for a particular procurement shall be as specified in the contract or order.

TABLE I. Dimensions and weights.

Size	Working surface dimensions (approximate)		Top thickness (minimum)	Depth overall (minimum)	Weight (minimum)
	Width, inches	Length, inches	inches	inches	pounds
1	2	4	5/8	5/8	1-1/2
2	3	5	3/4	3/4	2-3/4
3	6	6	7/8	7/8	8
4	8	8	7/8	7/8	10
5	10	14	3/4	4	35
6	14	18	3/4	4	70
7	18	24	3/4	4	150
8	24	48	3/4	4	400
9	30	84	1	6	1,300
10	36	72	1	6	1,400

FSC 3460

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Material. The lapping plates shall be made of fine grade gray iron or iron alloy castings.

Design. Lapping plates 8 by 8 inches and smaller shall be solid. Sizes greater than 8 by 8 inches shall include diagonal cross ribbing or other suitable means to insure maximum stability. Three point bearing support shall be provided at the junction of ribbing. Lapping plates 14 by 18 inches and larger shall be provided with handles, lifting holes or other suitable means for handling.

Castings. All castings shall be of uniform quality, free from defects, scale, and mismatching. In no event shall such processes as welding, peening, or filling with cold solders or metallic pastes be used for reclaiming any casting.

Grooves. Unless otherwise specified, the working surface shall be grooved to form a square or diamond pattern with the grooves either "V" shaped or straight sided. Plates 14 by 18 inches and over shall have grooves spaced approximately 1/2 inch center to center and be approximately 3/32 inch wide by 1/16 inch deep. Plates 10 by 14 inches and less shall have grooves approximately 1/8 inch center to center and approximately 1/32 inch wide by 1/32 inch deep. For the diamond pattern, the smaller angle between the grooves shall be 45 to 75 degrees.

Accuracy. The accuracy of the working surface shall be such that no two points on that surface vary with each other by more than 0.0002 inch per linear foot of space separating the two points.

Hardness. The work surface of lapping plates shall have a Brinell hardness of not less than 180.

Surface roughness. The maximum surface roughness of the working surface shall not exceed 32 microinches arithmetic average (AA) center line average. The four sides of the top shall not exceed 125 microinches AA.

Painting. All surfaces not machined shall be painted in accordance with the manufacturers standard painting procedure.

Marking. Lapping plates shall be marked in a permanent manner with the manufacturer's name or trademark.

Covers. All lapping plates shall be furnished with a properly fitted wood or other adequate material cover to protect the working surfaces and edges during shipping and storage.

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Measurement system. All dimensions in this description are given in the English system of measurement. These dimensions can be converted to the metric system equivalents by the use of ANSI/IEEE 268 for the Department of Defense use of FED-STD-376 for Civil Agency use.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table II that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with all requirements specified herein. Any sampling failing to meet the salient characteristics shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table II, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table II, the entire lot shall be considered defective and unsuitable for acceptance.

TABLE II. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance, number defective	Rejection, number defective
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	0	1
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6

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Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with ASTM D 3951. When specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with PPP-T-1150, level as specified.

Notes. The purchaser shall specify the quantity and size required for a specific procurement. When a particular pattern is desired, square or diamond, or a particular groove, V shaped or straight sided, purchaser should specify the pattern and groove required.

ANSI/IEEE 268 "Metric Practice" is available from American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.

ASTM D 3951 "Packaging, Commerical" is available from American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.

FED-STD-376 "Preferred Metric Units for General Use by the Federal Government" and Federal Specification PPP-T-1150 "Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery, Packaging, and Packing of" should be obtained from the procuring activity or as directed by the contracting officer.

Cross reference. The sizes listed in this CID are cross referenced to those listed in specification GGG-P-446 as shown in table III.

TABLE III. Cross reference.

GGG-P-446 size	A-A-51141 size	GGG-P-446 size	A-A-51141 size
2 x 4	1	14 x 18	6
3 x 5	2	18 x 24	7
6 x 6	3	24 x 48	8
8 x 8	4	30 x 84	9
10 x 14	5	36 x 72	10

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MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

DLA - IP

Project 3460-0092

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